

A commercial Roundup® formulation induced male germ cell apoptosis by promoting the expression of XAF1 in adult mice

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ABSTRACT

Roundup® is extensively used for weed control worldwide. Residues of this compound may lead to side effects of the male reproductive system. However, the toxic effects and mechanisms of Roundup® on male germ cells remain unclear. We aimed to investigate the apoptosis-inducing effects of Roundup® on mouse male germ cells and explore the role of a novel tumor suppressor XAF1 (X-linked inhibitor of apoptosis-associated factor 1) involved in this process. We demonstrated that Roundup® can impair spermatogenesis, decrease sperm motility and concentration, and increase the sperm deformity rate in mice. In addition, excessive apoptosis of germ cells accompanied by the overexpression of XAF1 occurred after Roundup® exposure both in vitro and in vivo. Furthermore, the low expression of XIAP (X-linked inhibitor of apoptosis) induced by Roundup® was inversely correlated with XAF1. Moreover, the knockdown of XAF1 attenuated germ cell apoptosis, improved XIAP expression and inhibited the activation of its downstream target proteins, caspase-3 and PARP, after Roundup® exposure. Taken together, our data indicated that XAF1 plays an important role in Roundup®-induced male germ cell apoptosis. The present study suggested that Roundup® exposure has potential negative implications on male reproductive health in mammals.

1. Introduction

Glyphosate-based herbicides (GBHs) are used worldwide in agriculture, forestry, and aquatic weed control (Benbrook, 2016). In addition, GBHs are spread on most eaten transgenic plants, modified to tolerate high levels of these compounds in their cells (Schütte et al., 2017; Owen, 2008). The most used commercial formulation of GBHs is Roundup®, which comprises mixtures of glyphosate and adjuvants such as polyoxyethylene tallowamine (POEA), to enhance the uptake and translocation of the active ingredient in plants (Brausch and Smith, 2007). Due to the large-scale use of GBHs and its accumulation in the environment and edible products, several major concerns have arisen in recent years about the harmful side effects of GBHs for soil and water quality, and plant, animal and human health (Van Bruggen et al., 2018; Bai and Ogbourne, 2016; EFSA, 2015; Guyton et al., 2015; Mesnage et al., 2015).

The testis is a sensitive target for xenobiotics (Main et al., 2010). The rapid decrease in male sperm quality worldwide in recent decades is considered related to environmental pollutants, including pesticides

(Barratt et al., 2017; Sifakis et al., 2017). It has been reported that GBHs could act as a chemical endocrine disruptor by disturbing the androgen/estrogen balance (Owagboriaye et al., 2017; Nardi et al., 2017). The variation in the reproductive hormone level could be explained by the changes in the number and function of Leydig cells and modification of steroidogenic acute regulatory protein (StAR) or aromatase levels and activity (Walsh et al., 2000; Cassault-Meyer et al., 2014; Defarge et al., 2016). Another study suggested that Roundup® may be inhibitory to the hypothalamic-pituitary axis, leading to reduction in the cyclic adenosine monophosphate (cAMP)/protein kinase A (PKA) pathway, StAR phosphorylation and corticosterone synthesis in the adrenal tissue (Pandey and Rudraiah, 2015). Recently, it was demonstrated that low concentrations of GBHs can cause complete loss of sperm motility in the yellowtail tetra fish *Astyanax lacustris* (Gonçalves et al., 2018). However, the adverse effects and mechanisms of GBHs in male germ cells remain unclear.

It was shown that spontaneous apoptosis of germ cells appears during normal spermatogenesis, and apoptosis plays an essential role in the potential spermatozoa output and guarantee of semen quality

Abbreviations: GBHs, glyphosate-based herbicides; XAF1, X-linked inhibitor of apoptosis-associated Factor 1; XIAP, X-linked inhibitor of apoptosis; TUNEL, terminal deoxynucleotidyl transferase dUTP nick end labeling; POEA, polyoxyethylene tallowamine

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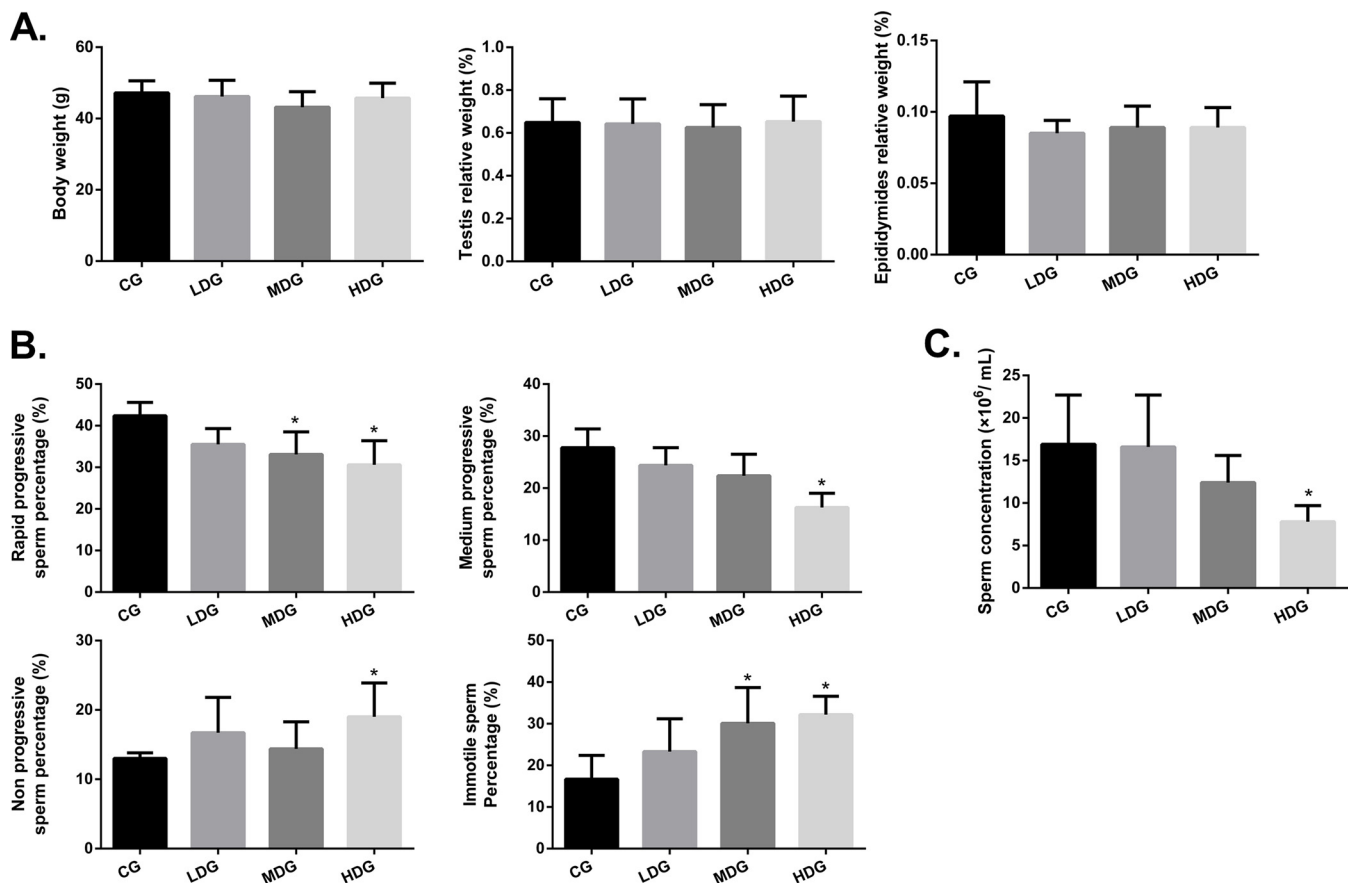


Fig. 1. Effects of Roundup® exposure for 35 days on body weight, relative weight of reproductive organs, epididymal sperm motility and concentration in mice. (A) No significant differences were found in all groups in body weight, testis relative weight and epididymis relative weight. (B) Sperm motility was decreased obviously after Roundup® exposure. (C) The epididymal sperm concentration was reduced after Roundup® exposure. The data are shown as the means $\pm$ SD. * $P < 0.05$, compared with the control group.

(Shaha et al., 2010; Mahfouz et al., 2009). However, extensive apoptosis of germ cells will create certain problems influencing normal spermatogenesis and may even lead to infertility (Liu et al., 2017), whereas the disturbance of spermatogenic cell apoptosis could improve male infertility (Kwon et al., 2004; Weikert et al., 2005). The aim of the present study, carried out in mice testis and a mouse spermatocyte cell line (GC-2), was to examine the role of apoptosis in the male reproductive toxicity of Roundup® and intracellular mechanisms involved in the apoptotic event.

2. Materials and methods

2.1. Reagents and animal treatment

The Roundup® formulation (Monsanto Co, St. Louis, MO, USA) contained 360 g/L of glyphosate [N-(phosphonomethyl)glycine] and 18% (w/v) POEA (surfactant). Solutions of the Roundup® formulation were prepared by the addition of appropriate volumes of distilled water. Roundup® containing 540 mg/kg bw/d, 180 mg/kg bw/d and 60 mg/kg bw/d glyphosate were selected as the “high”, “middle” and “low” doses, respectively, in this study, similar to previous studies (Jasper et al., 2012; Aitbali et al., 2018). Adult 8-week-old male Kunming mice approximately 34 g ($n = 32$) were obtained from the Experimental Animal Center of the Army Medical University. The animals were maintained under pathogen-free conditions with a 12-h light/12-h dark schedule and were fed with an autoclaved diet and water ad libitum. Thirty-two mice were randomly allocated into 4 groups ($n = 8$ per group): CG (control group), LDG (low-dose group, containing 60 mg/kg of glyphosate), MDG (middle-dose group,

containing 180 mg/kg glyphosate), and HDG (high-dose group, containing 540 mg/kg glyphosate). Animals were administered the indicated doses of Roundup® diluted in distilled water or with distilled water alone as a control each day by gavage. After treatment for one spermatogenesis cycle (35 days), the mice were anesthetized with 1% pentobarbital sodium and then were sacrificed. After removal of the tunica albuginea, the left testes were cut into two parts before being frozen in liquid nitrogen and stored at -80°C until RNA and protein extraction. The entire right testes were conserved overnight in Bouin's liquid and then were embedded in paraffin blocks prior to HE staining, TUNEL assay and immunofluorescence staining. All procedures were conducted in accordance with the guidelines for the Use and Care of Laboratory Animals and were approved by the Institutional animal care and use committee of Army Medical University.

2.2. Sperm motility and concentration analysis by the sperm class analyzer (SCA)

After anesthetization with 1% pentobarbital sodium, the sperm of the mice were isolated from the cauda epididymis, and the sperm motility and concentration parameters were detected as previously described (Jiang et al., 2016). Briefly, two sides of the cauda epididymis were transferred into a small dish with 150 μL of human tubal fluid (HTF) and were cut into pieces using iridectomy scissors to prepare the sperm suspension. Next, 5 μL of sperm sample was inserted into a Leja slide to analyze the sperm motility and concentration using the SCA system (Microptic S.L., Barcelona, Spain).

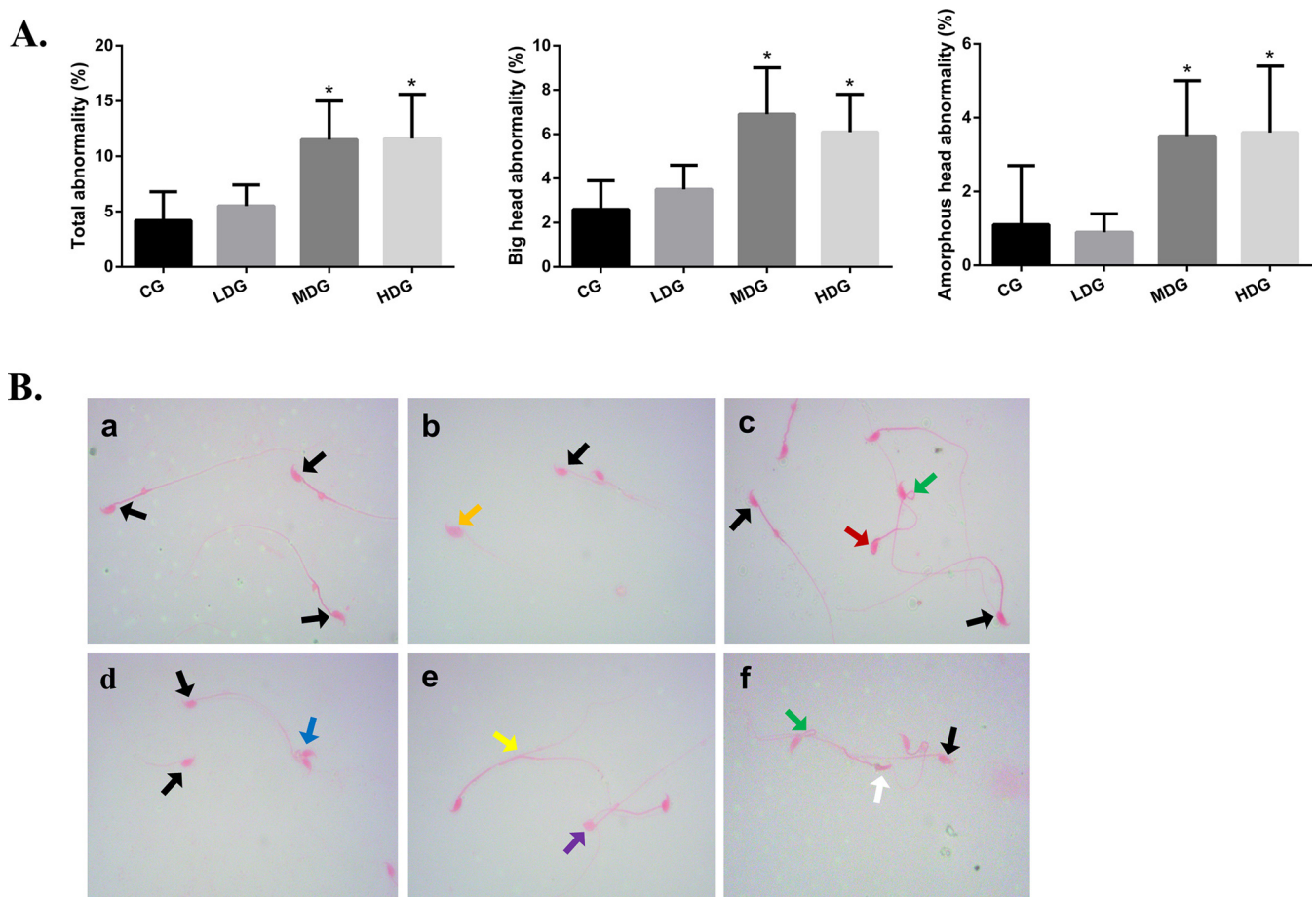


Fig. 2. Effects of Roundup® treatment for 35 days on mouse sperm morphology. (A) The sperm deformity rate was increased after Roundup® exposure. (B) Representative photographs of anomalous sperm morphology ($\times 400$ magnification). Normal sperm (Black arrow); Large-head sperm (Orange arrow); Non-hook head sperm (Red arrow); Coiled-tail sperm (Green arrow); Double-head sperm (Blue arrow); Double-tail sperm (Yellow arrow); Amorphous-head sperm (Purple arrow); Small-head sperm (White arrow). The data shown are the means $\pm$ SD. * $P < 0.05$, compared with the control group (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

2.3. Epididymal sperm morphology analysis

A 5- μ L droplet of epididymal sperm suspension was placed near the frosted end of the glass slide to prepare the sperm smear. After air drying, the samples were fixed with absolute ethyl alcohol for 15 min and then were stained with 2% eosin solution for 2 h. The morphology of the 300 sperm per mice and 5 mice in each group were observed randomly and classified into normal and different types of abnormal sperm under light microscopy (Olympus DP73, Tokyo, Japan). Next, the percentage of abnormal sperm was calculated to evaluate the effects of Roundup® on sperm morphology.

2.4. Hematoxylin and eosin (HE) staining

Paraffin-embedded testes tissue samples from the Roundup®-treated or untreated mice were used to prepare 5- μ m-thick sections. The sections were stained with HE and then were analyzed under a light microscope (Olympus DP73). The diameter of the seminiferous tubules was analyzed by ImageJ software and statistical analyzed using SPSS 16.0 software. At least 20 essentially round seminiferous tubules in five different testes were used for these measurements.

2.5. In situ detection of apoptotic cells by the TUNEL assay

Apoptosis of testicular samples was detected by terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) according to

the instructions of the in situ Dead End™ Colorimetric TUNEL System (Promega, USA).

2.6. Cell line and cell viability assays

Mouse GC-2 cells (an immortalized spermatocyte-derived cell line) was obtained from the American Type Culture Collection (ATCC, Rockville, MD, USA) and was cultured in DMEM high-glucose medium (HyClone, Logan, UT, USA) supplemented with 10% fetal bovine serum (Gibco BRL, Rockville, MD, USA) and incubated in a humidified atmosphere with 5% CO₂ at 37 °C. The viability of GC-2 cells treated with Roundup® was examined using the CCK8 assay according to the manufacturer's instructions (CCK-8, Dojindo, Japan). Cells were seeded in 96-well plates at a density of 3000 cells/well and then were divided into 4 groups: control group (containing 0 mM glyphosate), 0.04-mM group (containing 0.04 mM glyphosate), 0.08-mM group (containing 0.08 mM glyphosate) and 0.16-mM group (containing 0.16 mM glyphosate). After 18 h, the GC-2 cells were treated with various concentrations of Roundup® for 24 h or 48 h. All experiments were repeated three times.

2.7. Cell apoptosis analysis by flow cytometry (FCM)

To assess apoptosis, cells were determined by staining using an Annexin V-FITC/PI Apoptosis Detection kit (BD Biosciences, USA) according to the manufacturer's instructions and were analyzed using the

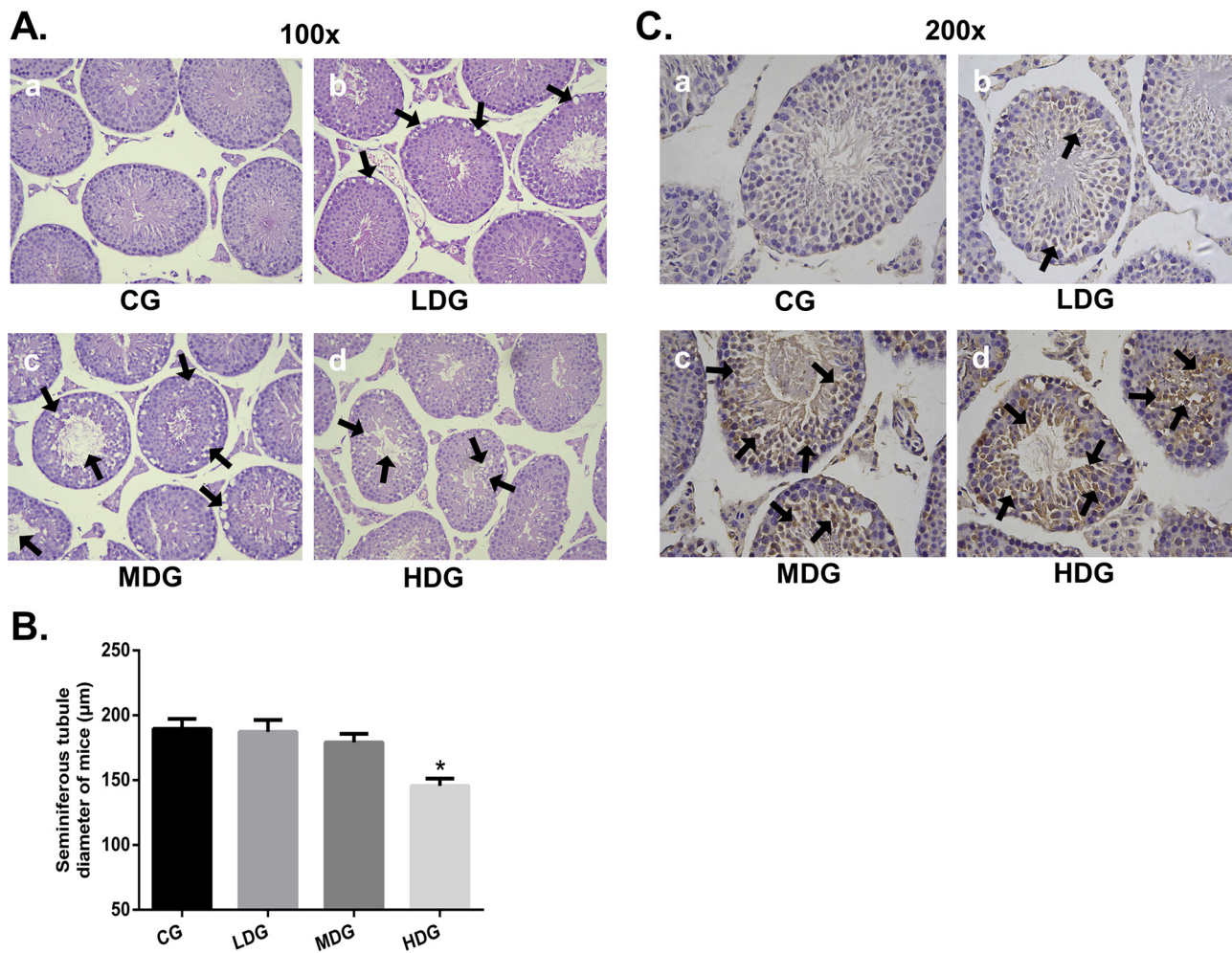


Fig. 3. Roundup® exposure results in histological lesions and induces apoptosis in the adult mouse testis. (A) Representative light micrographs of testes from the control and Roundup®-exposed mice ($\times 100$ magnification). a: The testis in CG presented a normal cellular architectural structure of the testicular tissue and an orderly arrangement of spermatogenic cells. b: Obvious vacuolization of the seminiferous tubule was observed in LDG. c: Mice in the MDG showed marked vacuolation of the seminiferous tubule, irregular arrangement of germ cells and a moderately depleted amount of germ cells. d: Severe testicular distortions characterized by degeneration of testicular tissue, shrinkage of the seminiferous tubule, a decrease in the diameter of seminiferous tubules, loosely arranged germ cells and alterations in the generation of sperm cells in HDG were observed. (B) Comparison of the seminiferous tubule diameter of mouse testes. (C) Spermatogenic cellular apoptosis after Roundup® exposure were detected by TUNEL staining ($\times 200$ magnification; The arrows indicate apoptotic cells). The data shown are the means $\pm$ SD. * $P < 0.05$, compared with the control group.

BD Accuri C6 FCM equipped with FlowJo 7.6 software.

2.8. RNA extraction and quantitative real-time PCR analysis

Total RNA was extracted from cells using TRIzol reagent (Invitrogen, Life Technologies Corporation, USA). cDNAs were obtained using the GoScript™ Reverse Transcription System Kit (Promega, Biotech Co., Ltd. USA). The mRNA level of XAF1 (X-linked inhibitor of apoptosis-associated factor 1) was detected using the CFX96 Touch™ Real-Time PCR Detection System (Bio-Rad Laboratories, Inc., USA) with SYBR Green PCR Master mix (Promega, USA). The primer sequences were synthesized by Sangon Biotech Co., Ltd. (Shanghai, China) and are listed as follows: XAF1, sense: 5'-GACCATGGAGGCTGACTTCC-3'/antisense: 5'-GGTGCTGTTGGCTTTCCTTG-3'; and β -actin, sense: 5'-GCTGTCCCTGTATGCCTCT-3'/antisense: 5'-GGTCTTTACGGATGTCAACG-3'.

2.9. Immunofluorescence staining

Cells cultured on glass coverslips were treated with Roundup® for 48 h, washed with PBS, fixed with 4% paraformaldehyde for 10 min,

and permeabilized with 0.5% Triton X-100 in PBS for 5 min. Meanwhile, paraffin-embedded testis sections were deparaffinized in xylene and ethanol. Thereafter, endogenous peroxidase activity was blocked by incubating the slides for 20 min at room temperature in 3% hydrogen peroxide diluted in methanol. The slides were then washed with PBS and incubated for 1 h in 5% bovine serum albumin to prevent nonspecific binding. Next, the coverslips and slides were incubated with anti-XAF1 antibody (ab17204; Abcam, UK; 1:100) and stained with DyLight 549-conjugated secondary antibody (A23320; Abbkine Scientific Co., Ltd, China; 1:400). The nuclei were fluorescently labeled with DAPI. The coverslips were then washed and mounted on glass slides. Fluorescent images were obtained using a confocal microscope (Zeiss LSM 800; Carl Zeiss, Germany).

2.10. Transient transfection with siRNA

The sequences of the XAF1 and negative control siRNA were designed and synthesized, and the knockdown efficiency was evaluated at 48 h after transfection by real-time PCR. The cells were transfected with XAF1 siRNA (siXAF1) or negative control siRNA (NC) using Lipofectamine 2000 (Invitrogen Carlsbad, CA, USA) according to the

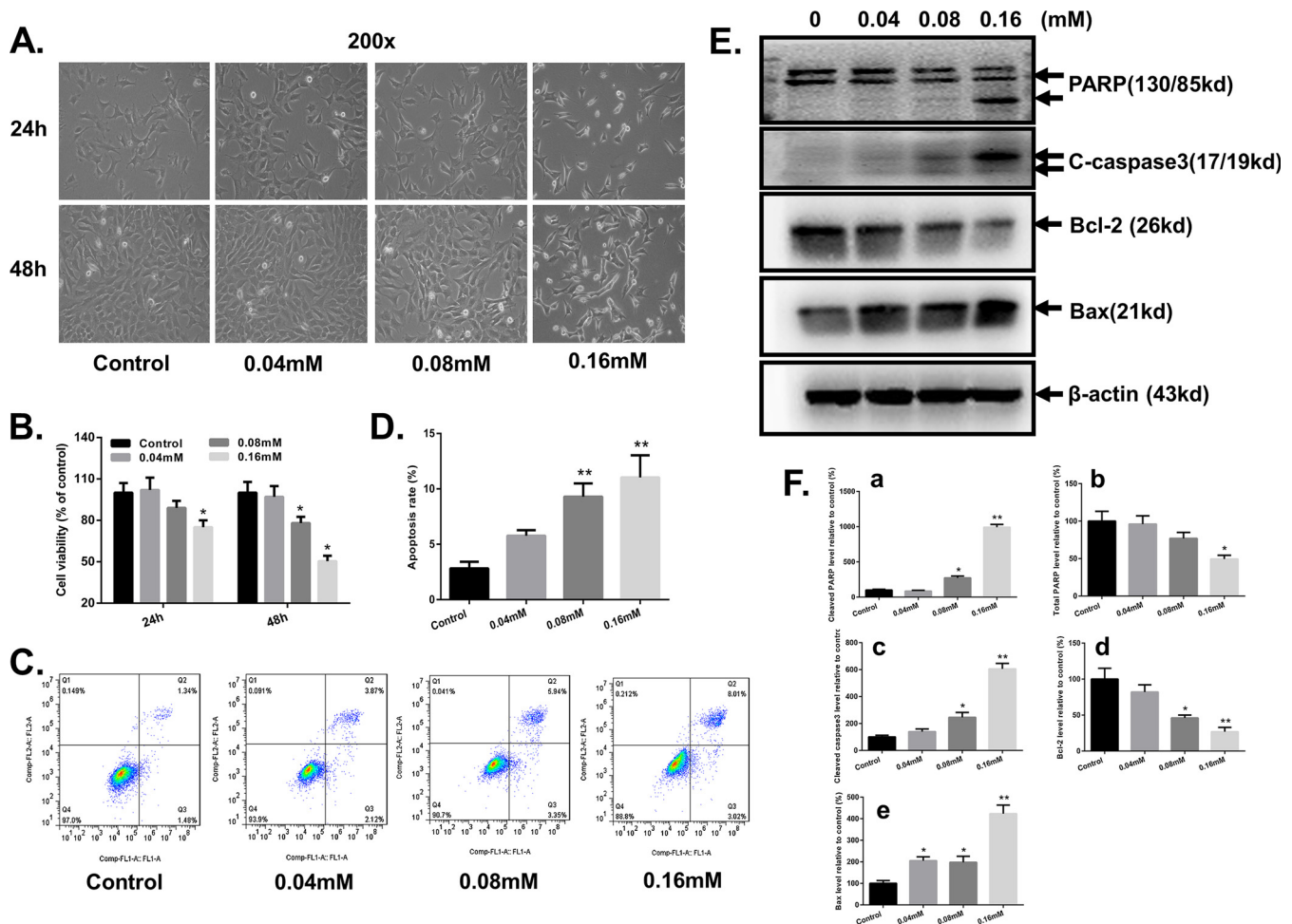


Fig. 4. Roundup® inhibits growth and induces apoptosis in GC-2 cells. (A) GC2 cells were treated with vehicle alone or specified concentrations of Roundup® (containing 0.04 mM, 0.08 mM and 0.16 mM glyphosate) for 24 h or 48 h. Representative micrographs of different groups were observed ($\times 200$ magnification). (B) Roundup® reduced GC-2 cell viability in a dose- and time-dependent manner. (C) After Roundup® exposure for 48 h, the apoptosis rate of GC-2 cells was determined by FCM. (D) Quantitative analysis of the FCM results. (E) After Roundup® exposure for 48 h, the expression levels of apoptosis-related proteins were detected by WB. (F) Quantitative analysis of the WB results. All the data represented the means $\pm$ SD of three independent experiments. * $P < 0.05$, ** $P < 0.01$, compared with the control.

manufacturer's instructions. The siRNA sequences used were as follows: siRNA (siXAF1): 5'-UGCAGCUUAGCAAUCUGGAUGUCCA-3'; and negative control siRNA (NC): 5'-UUCUCCGAACGUGUCACGUTT-3'.

2.11. Western blot analysis (WB)

Total protein was extracted from cell lines and the mice testis. The protein concentrations were determined using the Enhanced BCA Protein Assay Kit (P0010 Beyotime, Jiangsu, China). The protein samples (60 μ g) were separated by 10% or 12% SDS-PAGE and then were transferred to equilibrated polyvinylidene fluoride membranes for 3 h at 80 V. After blocking with 5% dehydrated skim milk, the membranes were incubated with primary antibodies against XAF1 (ab17204; Abcam, UK; 1:1000), PARP (ab32138; Abcam, UK; 1:3000), Bcl-2 (ab182858; Abcam, UK; 1:2000), Bax (ab32503; Abcam, UK; 1:4000), cleaved caspase-3 (9661; Cell Signaling Technology, USA; 1:1000), XIAP (X-linked inhibitor of apoptosis; ab21278; Abcam, UK; 1:1000) and β -actin (AA128; Beyotime, China; 1:1000) overnight at 4 $^{\circ}$ C, followed by incubation with the HRP-conjugated goat anti-mouse secondary antibody (A0216; Beyotime, China; 1:1000) or HRP conjugated goat anti-rabbit second antibody (A0208; Beyotime, China; 1:1000) for 1.5 h at room temperature. The proteins were detected using the enhanced chemiluminescence kit (Millipore, Germany) using an automated western blot imaging system (FUSION FX Spectra; VILBER,

France). The WB images were quantified using ImageJ software and were statistical analyzed using SPSS 16.0 software.

2.12. Statistical analysis

The results were expressed as the means $\pm$ SD compared using the F-test and q-test and SPSS 16.0 statistical software. $P < 0.05$ was considered significantly different.

3. Results

3.1. Roundup® shows no significant effects on the mouse body weight and relative weight of reproductive organs

In this study, daily exposure to Roundup® for 35 days caused no overt signs of toxicity in male adult mice. As shown in Fig. 1A, the body weights in the CG, LDG, MDG and HDG groups were 47.19 ± 3.38 g, 46.15 ± 4.55 g, 43.16 ± 4.38 g and 45.70 ± 4.21 g, respectively. Meanwhile, the relative testis weights in the CG, LDG, MDG and HDG groups were $0.65 \pm 0.11\%$, $0.64 \pm 0.10\%$, $0.63 \pm 0.13\%$ and $0.65 \pm 0.12\%$, respectively. The relative weights of the epididymides in the CG, LDG, MDG and HDG groups were $0.10 \pm 0.02\%$, $0.08 \pm 0.01\%$, $0.09 \pm 0.01\%$ and $0.09 \pm 0.01\%$, respectively. Together, these results demonstrated that Roundup® exposure showed no

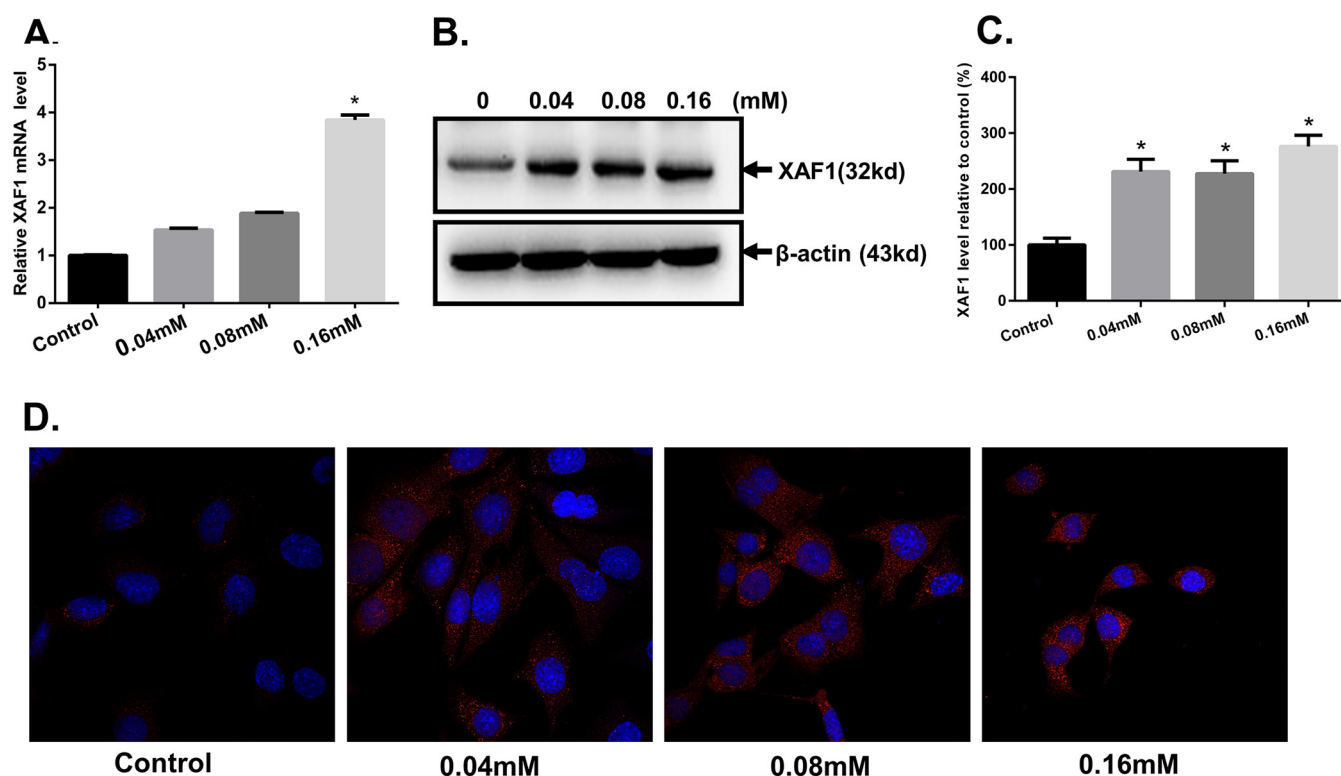


Fig. 5. Roundup® induces the expression of XAF1 at both the mRNA and protein levels in GC-2 cells. (A) The XAF1 mRNA levels were determined by q-PCR after treatment with Roundup® at the indicated doses for 48 h. (B) The protein levels of XAF1 after Roundup® treatment were determined by WB. (C) Quantitative analysis of the WB results. (D) Immunofluorescence analysis of XAF1 levels in Roundup®-treated GC-2 cells. Each experiment was repeated three times. The data shown are the means $\pm$ SD. * $P < 0.05$, compared with the control group.

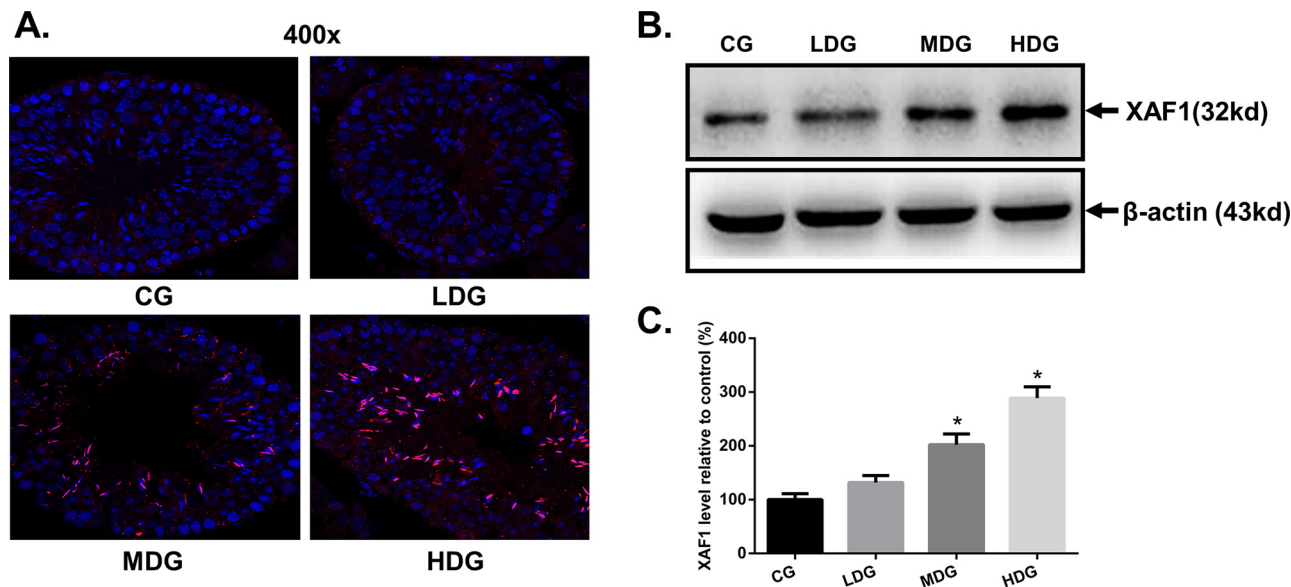


Fig. 6. Roundup® promotes XAF1 expression in vivo. (A) Immunofluorescence analysis of XAF1 expression after Roundup® exposure in mouse testes. (B) The protein levels of XAF1 originated from the testis of Roundup®-exposed mice were detected by WB. (C) Quantitative analysis of the WB results. The data shown are the means $\pm$ SD. * $P < 0.05$, compared with the control group.

significant effects on the body weight and relative weights of reproductive organs in mice.

3.2. Roundup® decreases the epididymal sperm motility and concentration in mice

As shown in Fig. 1B, the rapid progressive sperm percentages in the

MDG and HDG groups were $33.1 \pm 5.4\%$ and $30.6 \pm 5.8\%$, respectively, which were decreased significantly compared with that in the CG group at $42.4 \pm 3.2\%$. Meanwhile, the immotile sperm percentages of the LDG, MDG and HDG groups were $23.3 \pm 7.9\%$, $30.1 \pm 8.6\%$ and $32.2 \pm 4.4\%$, respectively, which were increased compared with that of CG at $17.6 \pm 5.7\%$. On the other hand, the sperm concentration in the HDG group ($7.8 \times 10^6/\text{mL}$) was significantly lower than that in the

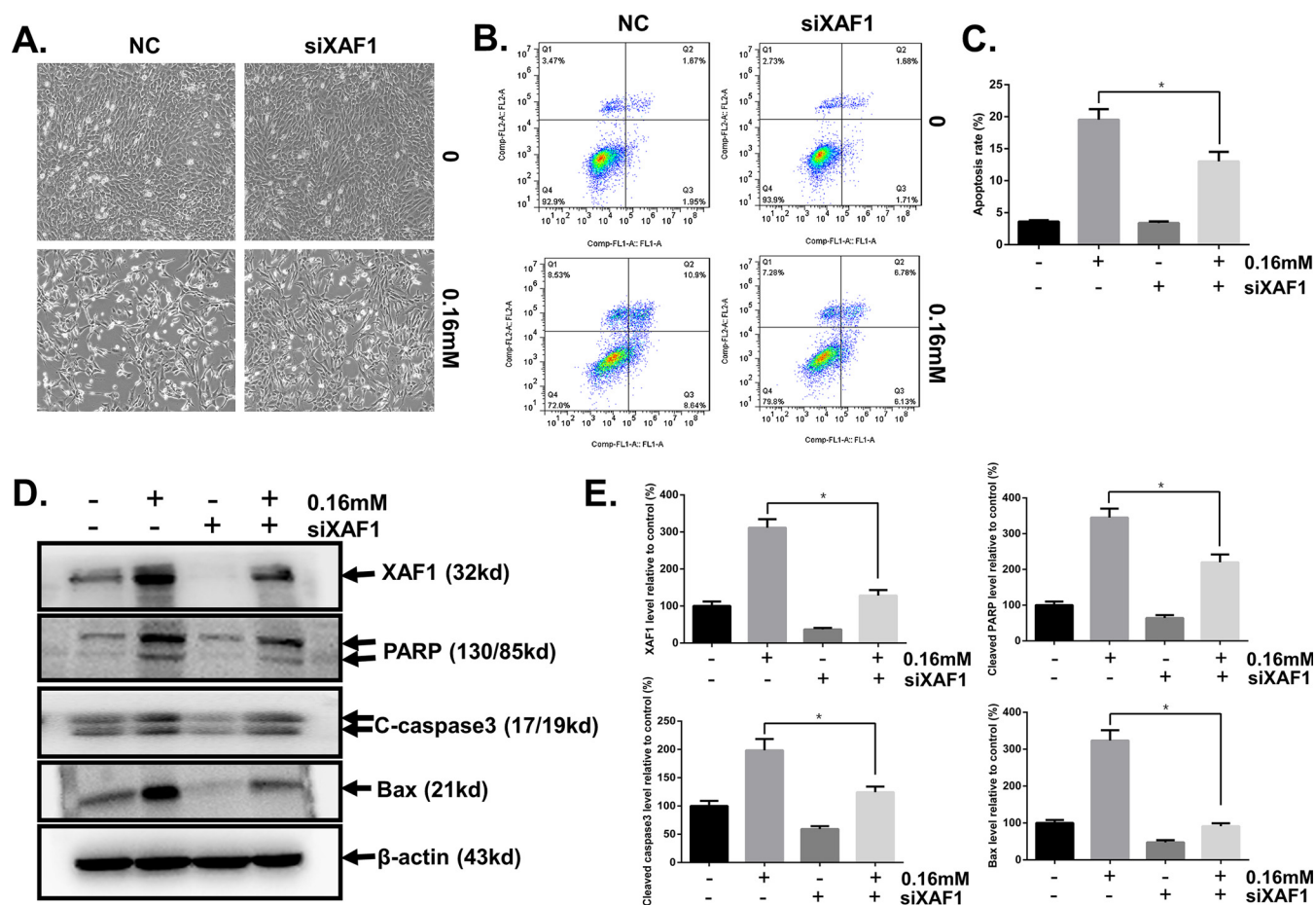


Fig. 7. XAF1 mediates Roundup®-induced apoptosis in GC-2 cells. (A) Photomicrographs of GC-2 cells treated with Roundup® or vehicle for 48 h after transfection with control siRNA or XAF1 siRNA. (B) The apoptosis rates were analyzed using flow cytometry stained by annexin V-FITC/PI after transfection and Roundup® treatment. (C) Quantitative analysis of the FCM results. (D) The expression levels of XAF1 and other apoptosis-related proteins were detected by WB after XAF1 siRNA transfection and Roundup® treatment. (E) Quantitative analysis of the WB results. All the data represented the means $\pm$ SD of three independent experiments. * $P < 0.05$, compared with the control.

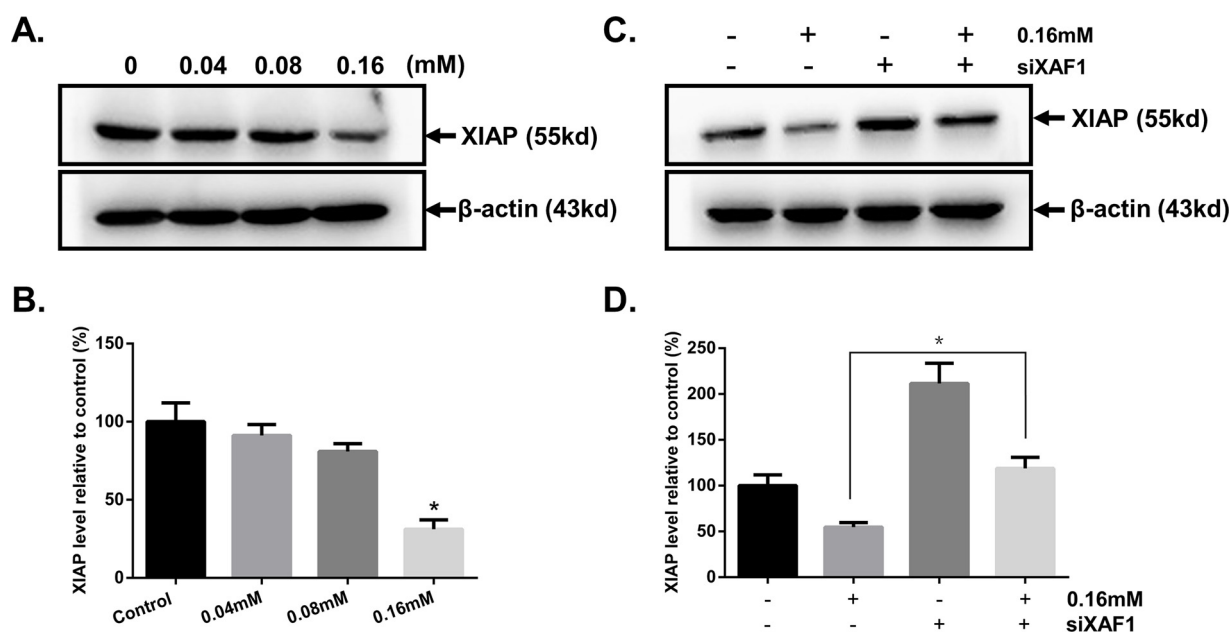


Fig. 8. XAF1 suppresses XIAP expression after Roundup® exposure in GC-2 cells. (A) The protein level of XIAP was detected in GC-2 cells after Roundup® exposure for 48 h by WB. (B) Quantitative analysis of (A). (C) The expression level of XIAP in GC-2 cells was measured after transfection with siRNA targeting XAF1 and treatment with Roundup®. (D) Quantitative analysis of (C). The data shown are the means $\pm$ SD. * $P < 0.05$, compared with the control.

CG group ($16.9 \times 10^6/\text{mL}$, Fig. 1C). These data indicated that Roundup® can decrease sperm vitality and sperm density in the mouse epididymis.

3.3. Roundup® increases the sperm deformity rate in mice

After treatment with Roundup® for 35 d, the total abnormalities of deformed sperm in the MDG ($11.5 \pm 3.5\%$) and HDG ($11.6 \pm 4.0\%$) groups were increased significantly compared with that in the CG group ($4.2 \pm 2.6\%$). Additionally, we found that the deformity rate of large-head sperm was $2.6 \pm 1.3\%$ in the CG group but $3.5 \pm 2.1\%$, $6.9 \pm 2.1\%$ and $6.1 \pm 1.7\%$ in the LDG, MDG and HDG groups, respectively. The percentage of amorphous-head sperm was $1.1 \pm 1.4\%$ in the CG group and $0.9 \pm 0.5\%$, $3.5 \pm 1.5\%$ and $3.6 \pm 1.8\%$ in the LDG, MDG and HDG groups, respectively (Fig. 2A). These results suggested that Roundup® causes abnormal sperm morphology, mainly in the head region. The representative photographs of anomalous sperm are shown in Fig. 2B.

3.4. Roundup® results in morphological changes and germ cell apoptosis in the mouse testis

To assess the adverse effects of Roundup® in vivo, we observed the morphology of mouse testes after Roundup® exposure for 35 days by optical microscopy. As shown in Fig. 3A, the testis in the CG group presented a normal cellular architectural structure of the testicular tissue and an orderly arrangement of spermatogenic cells. Progressive degenerative lesions were observed in mouse testes exposed to different concentrations of Roundup®. Obvious vacuolization of the seminiferous tubule was observed in the LDG group. Mice in the MDG group showed marked vacuolation of the seminiferous tubule, irregular arrangement of germ cells and a moderately depleted amount of germ cells. Severe testicular distortions characterized by degeneration of testicular tissue, shrinkage of the seminiferous tubule, a decrease in the diameter of seminiferous tubules, loosely arranged germ cells and alterations in the generation of sperm cells in the HDG group were observed. Additionally, we found that the diameter of the seminiferous tubule in the HDG group was significantly reduced compared with that in the CG group (Fig. 3B). These data suggested that Roundup® can impair spermatogenesis in the mouse testis.

Next, we detected whether Roundup® can induce spermatogenic cellular apoptosis in the mouse testis by the TUNEL assay. Apoptotic cells were stained with brown color. As shown in Fig. 3C, the CG group showed a normal apoptotic profile, with few positively stained apoptotic cells in the seminiferous tubules. By contrast, many positive germ cells, mainly spermatocytes and spermatids, appeared in the MDG and HDG groups.

3.5. Roundup® inhibits the viability and induces apoptosis in GC-2 cells

The results of the CCK8 assay revealed that the viability of GC-2 cells treated with Roundup® was decreased in a dose- and time-dependent manner (Fig. 4B). The representative photographs are shown in Fig. 4A. The apoptotic rates were $2.82 \pm 0.61\%$, $5.77 \pm 0.53\%$, $9.29 \pm 1.20\%$, and $11.03 \pm 2.11\%$ in the control group, 0.04-mM group, 0.08-mM group and 0.16-mM group, respectively (Fig. 4C–D). As shown in Fig. 4E, Roundup® induced the cleavage of caspase-3 and PARP. Meanwhile, the decreased level of the anti-apoptotic protein Bcl-2 and increased level of the pro-apoptotic protein Bax were found after Roundup® exposure for 48 h. The results of the densitometry quantification of the bands for western blot experiments are shown in Fig. 4F. These results indicated that Roundup® treatment can induce GC-2 cell apoptosis, a finding that is consistent with the in vivo results.

3.6. Roundup® enhances the expression of XAF1 both in vitro and in vivo

As a pro-apoptosis factor, XAF1 was selected to assess the change in

the expression level in GC-2 cells after exposure to Roundup®. As shown in Fig. 5A, the mRNA levels of XAF1 were upregulated in the Roundup®-treated groups, especially in the 0.16-mM group. From the WB and grayscale analysis results, the protein levels of XAF1 were also greatly elevated in GC-2 cells exposed to Roundup® (Fig. 5B–C). The enhanced protein expression of XAF1 after Roundup® exposure was also confirmed by immunofluorescence analysis (Fig. 5D).

To determine whether Roundup® can promote XAF1 expression in vivo, we detected the protein levels of XAF1 in the mouse testes after Roundup® exposure. Both the immunofluorescence assay and WB results indicated that Roundup® can promote the expression of XAF1 in the mouse testes obviously. In addition, the fluorescence signals of XAF1 in the Roundup®-treated groups were markedly localized in spermatocytes and spermatids, especially in the late spermatids, a finding consistent with the TUNEL results above (Fig. 6A–B). The results of the densitometry quantification of the bands for the WB in vivo study are shown in Fig. 6C.

3.7. XAF1 mediates Roundup®-induced apoptosis in GC-2 cells

Based on the above results, we speculated that XAF1 may be involved in Roundup®-induced germ cell apoptosis. To verify the hypothesis, we analyzed the apoptosis of GC-2 by FCM and WB after transfection by siXAF1 or NC to downregulate its expression. After the administration of Roundup® for 48 h, the apoptosis rate of the NC + Roundup® group was $19.5 \pm 0.53\%$, whereas that of the siXAF1 + Roundup® group was $12.9 \pm 0.66\%$ (Fig. 7B). The results were from three independent experiments. Representative photomicrographs and the quantitative data of FCM are shown in Fig. 7A and C. WB results showed that transfection specifically with siXAF1 can partly attenuate germ cell apoptosis, reduce the expression of Bax and inhibit the activation of caspase-3 and PARP after Roundup® exposure (Fig. 7D). The quantitative data of WB are shown in Fig. 7E. These results indicated that XAF1 mediates Roundup®-induced apoptosis in GC-2 cells.

3.8. XAF1 inhibits XIAP expression after Roundup® exposure

To explore the mechanisms by which XAF1 induce germ cell apoptosis after Roundup® exposure, we first detected the protein level of XIAP, which is an inhibitor of apoptosis. As shown in Fig. 8A–B, the protein levels of XIAP were decreased obviously in a dose-dependent manner after Roundup® exposure. Furthermore, XIAP levels were enhanced after knockdown of the expression of XAF1 specifically. Meanwhile, the lack of XAF1 can partly recover Roundup®-induced XIAP low expression (Fig. 8C–D).

4. Discussion

In the present study, we noticed the obvious pathological changes of the seminiferous tubules and degraded sperm quality, including sperm motility, sperm density and sperm morphology, after Roundup® exposure. These findings indicated the toxicity of Roundup® in spermatogenesis. Spermatogenesis is a complex biological process and the transition through different steps is tightly regulated (Gunes et al., 2015). A defect at any of these steps may lead to spermatogenesis dysfunction. Among several steps important in this process, quality control is one of the most important events. Apoptosis is the best-known quality-control mechanism in the testis (Shukla et al., 2012). Additionally, the number of apoptotic germ cells was reported to be higher in testicular biopsies from infertile men than that from fertile men (Said et al., 2004). Furthermore, sperm caspases were more activated in patients with infertility than in healthy men (Taylor et al., 2004). In addition to the physiological factors, external disturbances such as irradiation or exposure to toxicants can also result in the massive apoptosis of testicular germ cells (Calogero et al., 2011). From our study, the

positively stained apoptotic cells in the Roundup®-treated groups were mainly spermatocytes and spermatids and were increased obviously compared with those in the control group. Additionally, caspase-3 was activated obviously in GC-2 cells after Roundup® exposure, which is a sensitive endpoint for apoptosis. Meanwhile, a decreased level of the anti-apoptotic protein Bcl-2 and an increased level of the pro-apoptotic protein Bax were found after Roundup® exposure in GC-2 cells. These results suggested that excessive apoptosis of germ cells plays an important role in Roundup®-induced reproductive disorders, indicating a specific pathway is activated when the testicular environment cannot support spermatogenesis. However, the elements that control this process have not been identified.

XAF1 is a novel proapoptosis gene that has gained importance as a candidate tumor suppressor because of its apoptosis-inducing effect (Huang et al., 2010). To our knowledge, most of the studies regarding XAF1 were conducted in tumor cells, while studies concerning XAF1 that were performed in normal cells or healthy tissues are rare (Plenchette et al., 2007). Because XAF1 is ubiquitously expressed in normal tissues, including testicular tissue, we speculate that XAF1 may participate in the apoptosis events of germ cells. To understand whether XAF1 was involved in Roundup®-induced germ cell apoptosis, we detected the expression level of XAF1 and investigated the role and mechanism of XAF1 in Roundup®-induced excessive apoptosis in germ cells by knockdown experiments. From our results, the expression of XAF1 was elevated significantly after Roundup® exposure, which is accompanied by the increase in apoptotic germ cells. In addition, the knockdown of XAF1 expression partially inhibited the apoptosis of GC-2 cells induced by Roundup® and suppressed the activation of PARP and caspase-3 after Roundup® exposure. These data suggested that XAF1 plays an important role in conducting germ cell apoptosis. XAF1 may be a candidate indicator of Roundup®-induced apoptosis. Many studies have been performed to identify strategies to recover environment stress-damaged spermatogenesis and treat infertility. It was reported that the disturbance of spermatogenic cell apoptosis could improve male infertility (Kwon et al., 2004; Weikert et al., 2005). From this point of view, XAF1 may serve as a potential target for gene therapy to treat or at least relieve Roundup®-induced spermatogenesis damage.

XIAP, the most potent member of the inhibitor of apoptosis (IAP) family, inhibits caspases and blocks the apoptotic pathway (Salvesen and Duckett, 2002). Thus, downregulation of XIAP is recognized as an efficient anti-cancer approach (Li et al., 2013). XAF1 was originally identified as an XIAP-interacting protein that could bind and interfere with the anticaspase function of XIAP by sequestering XIAP protein to the nucleus (Liston et al., 2001). It was reported that XAF1 can bind to the RING domain of XIAP and activate the E3 ubiquitin ligase activity of XIAP that targets survivin for proteasomal degradation (Galbán and Duckett, 2010; Tse et al., 2012). Moreover, the overexpression of XAF1 could suppress XIAP expression, inhibit invasion and promote apoptosis (Zhao et al., 2015). However, XAF1 has been reported to evoke an apoptotic effect in XIAP^{-/-} cells to the extent comparable to that in XIAP^{+/+} cells, indicating that its apoptosis-promoting function is not solely dependent on the XIAP-interfering activity (Xia et al., 2006; Chung et al., 2007). To further explore the possible molecular mechanism of XAF1 in Roundup®-induced germ cell apoptosis, we detected the levels of XIAP after Roundup® exposure. At the same time, the knockdown experiment was performed to examine the expression of XIAP in XAF1-knockdown GC-2 cells. From the results, we found that the expression of XIAP was decreased obviously after Roundup® exposure and was inversely correlated with the expression of XAF1. Moreover, the knockdown of XAF1 specifically promoted the expression of XIAP and increased Roundup®-induced XIAP low expression. These results indicated that XAF1 induces male germ cell apoptosis by repressing the expression of XIAP. Because of the important roles of XAF1 in Roundup®-induced germ cell apoptosis, the transcriptional activation mechanisms controlling XAF1 expression after Roundup® exposure must be identified in future investigations.

It is necessary to point out that the commercial formulation of Roundup® is a mixture of glyphosate with varying and unspecified surfactants. These compounds are considered as inert but may also be toxic. In this study, we only detected the reproductive toxicity and mechanism of one commercial formula, which does not mean that all the commercial formulations of Roundup® are toxic in male reproduction. Therefore, it is necessary to evaluate the reproductive toxicity of Roundup® with various commercial formulations. Thus far, the potential toxicity of Roundup® in mammals remains controversial. In fact, some formulations have been identified in GBHs and are even more toxic than the active ingredient itself (Janssens and Stoks, 2017). From the new evaluation, glyphosate does not show carcinogenic or mutagenic properties or reproductive or embryonic developmental toxicities in laboratory animals. The new evaluation indicates that the measured toxicity of some GBHs is the result of the co-formulants (e.g., tallowamines used as surfactants) (Germany Rapporteur Member State, 2015). Therefore, we speculated that the male reproductive toxicity observed in this study may not directly result from glyphosate alone but from the surfactant or other mixtures. Therefore, additional studies concerning the toxicity of glyphosate alone, surfactant alone and the commercial formulation should be performed to ascertain the component(s) responsible for the reproductive toxicity.

5. Conclusions

The present study confirmed the male reproductive toxicity of Roundup® and demonstrated that treatment with Roundup® could induce aberrant apoptosis by promoting XAF1 expression and suppressing XIAP expression in germ cells. Our data preliminarily identified the significant role of XAF1 in Roundup®-induced germ cell apoptosis and suggested that XAF1 may serve as a trigger of apoptosis after Roundup® treatment.

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Conflict of interest

All authors declare that there are no conflicts of interest.

Transparency document

The Transparency document associated with this article can be found in the online version.

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